THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY

COE | DSAI
INNOVATING INTELLIGENCE

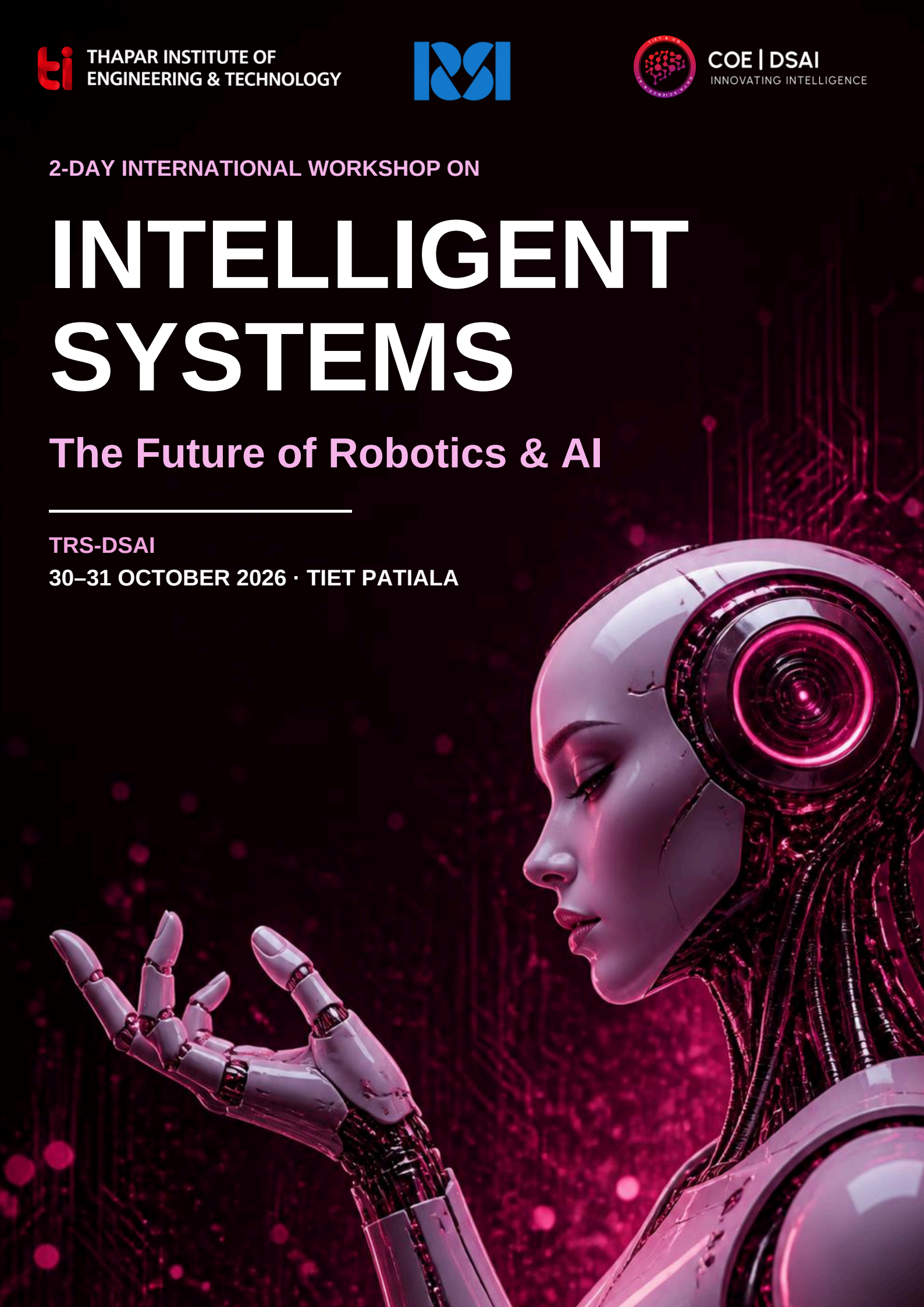
2-DAY INTERNATIONAL WORKSHOP ON

INTELLIGENT SYSTEMS

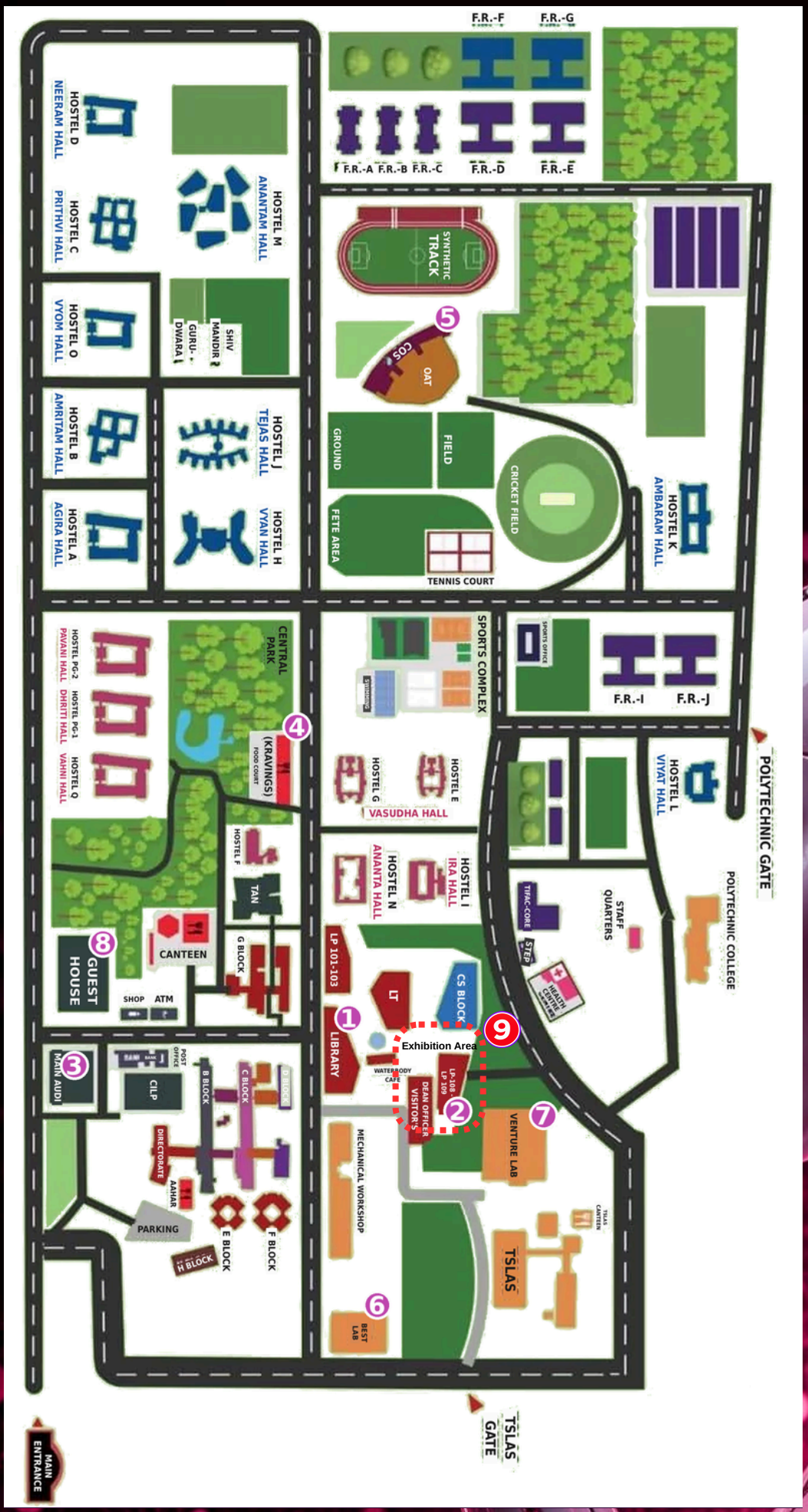
The Future of Robotics & AI

TRS-DSAI

30–31 OCTOBER 2026 · TIET PATIALA



TIET CAMPUS MAP



KEY LOCATIONS

1. Nava Nalanda Library
2. LP-108 & LP-109
3. Main Auditorium
4. Kravings
5. COS Complex
6. BEST Lab
7. Venture Lab
8. Guest House
9. Exhibition Area

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ABOUT THE WORKSHOP

TRS–DSAI Two-Day International Workshop on Intelligent Systems: The Future of Robotics and AI will be held on October 30–31, 2026 at Thapar Institute of Engineering & Technology (TIET), Patiala, India.

The workshop aims to bring together distinguished academicians, researchers, industry experts, startups, and students from across India and abroad to discuss recent advances and emerging trends in robotics, artificial intelligence, autonomous systems, and intelligent technologies. It will serve as a premier platform for fostering interdisciplinary collaborations, knowledge exchange, and academia–industry partnerships.

WORKSHOP THEMES:

- Robotics and Autonomous Systems
- Healthcare and Medical Robotics
- Digital Twins and Simulation
- Artificial Intelligence & Machine Learning
- Multi-Agent Systems

We look forward to welcoming you to TIET, Patiala, for an engaging and impactful event.

About The Workshop

ABOUT TRS & DSAI (CoE)

The Robotics Society (TRS)

TRS is an Indian professional society that promotes robotics research, education, innovation and collaboration among researchers, academicians, industry professionals and students. It organizes the international conference Advances in Robotics (AIR) every two years, along with workshops, faculty development programmes, and student chapters.

Key Activities

- Robotics research and development
- Conferences, workshops and FDPs
- Webinars and training programmes
- Robotics student chapters
- Academic–industry collaboration
- Newsletters and research outreach

Advances in Robotics (AIR), the flagship international conference of TRS, is held every two years. Previous editions were hosted at DRDO, Pune (2013), BITS Pilani, Goa (2015), IIT Delhi (2017), IIT Madras (2019), IIT Kanpur (2021, online), IIT Ropar (2023) and IIT Jodhpur (2025). The upcoming AIR 2027 will be hosted by IIT Gandhinagar in July 2027.

TIET–UQ Centre of Excellence in DSAI

The Centre of Excellence in Data Science and Artificial Intelligence (CoDSAI) is a landmark collaboration between Thapar Institute of Engineering & Technology, India, and The University of Queensland, Australia. It unites leading researchers from both institutions to drive high-impact, interdisciplinary innovation in AI and Data Science, and fosters talent development through research internships.

Strategic pillars: Research & Innovation • Education & Training • Industry Collaboration & Outreach

Thematic areas: Health & Medicine • Behavioural Science • Energy & Environment • Cybersecurity • Sustainable Infrastructure

ORGANIZING COMMITTEE



Prof. Padmakumar Nair
Vice Chancellor, TIET Patiala
Chief Patron



Prof. Ajay Batish
Pro Vice Chancellor, TIET Patiala
Patron



Shri. Alok Mukherjee
President, TRS
Patron



Prof. Inderveer Chana
Joint Director DSAI, TIET Patiala
Workshop Chair



Prof. Tim Miller
Joint Director DSAI, TIET Patiala
Workshop Chair



Prof. Ashish Singla
Professor, TIET Patiala
Convener



Prof. Sachin Kansal
Associate Professor, TIET Patiala
Co-Convener

SUPPORTING TEAM



Dr. Subhash Pratap
MED, TIET Patiala
Media & Outreach Team



Prof. Anju Bala
CSED, TIET Patiala
Hospitality Team



Prof. Dheeraj Gupta
MED, TIET Patiala
Event Management Team

CORE TEAM



Mr. Shavinder Singh
MED, TIET Patiala



Mr. Anikait Kaushik
MED, TIET Patiala



Mr. Keshav Chhabra
MED, TIET Patiala



Mr. Shreyas Patel
MED, IIT Ropar



Ms. Sakshi Dhiman
CSED, TIET Patiala



Ms. Diya Jain
CSED, TIET Patiala

VOLUNTEERING TEAM

Ratan Kapur	Priyanshu Sachdeva	Dhruv Gupta
Rudhratej Singh	Anieruddh Kandpal	Anvi Bhatia
Yukti Bhatia	Prisha Verma	Rishab Kumar Jha
Bawa Singh	Manish Jain	Rushil Kapur
Chhaya Chauhan	Dhriti Dhall	Rishab Singla
Vinayak Bansal	Aayush Sawhney	Kratika Gupta
Medhansh Negi	Agamvir Singh	Mehakpreet Kaur
Ritik Raj Singh	Raxit Verma	Hitanshi Dhingra
Harshit Soni	Prashaant Sharma	Mitul Patel
Ishaan Shishodia	Pratishtha Sharma	Devam Aggarwal
Omkar Kalyan	Armaan Gupta	Yashaswini
Rehan Babu	Jasmeet Singh	Anmol Singh

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Padmakumar Nair
Vice Chancellor, TIET Patiala
Chief Patron

MESSAGE

Thapar Institute of Engineering and Technology has, for more than seven decades, stood for academic excellence, integrity and a deep commitment to nation-building. Our strength lies in our people: dedicated faculty, curious students, committed staff and an accomplished alumni community spread across the world. As the landscape of education and work changes rapidly, we remain focused on nurturing graduates who think critically, act ethically and lead with purpose. Through interdisciplinary learning, impactful research, global partnerships and close engagement with industry and society, the Institute continues to grow as a vibrant centre of knowledge and innovation. I invite every member of the Thapar family to aspire high and contribute to a sustainable future.

ABOUT THE SPEAKER

Prof. Padmakumar Nair is the Vice Chancellor of Thapar Institute of Engineering and Technology (TIET), Patiala. He joined TIET in July 2013, where he led the LM Thapar School of Management and played a key role in establishing the Thapar School of Liberal Arts and Sciences. He holds PhDs from the University of Twente and the University of Tokyo, as well as an MBA from Heriot-Watt University. He is passionate about developing and leading sustainable enterprises.

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Ajay Batish
Pro Vice Chancellor, TIET Patiala
Patron

MESSAGE

At Thapar Institute of Engineering and Technology, our aim is to offer an education that combines strong fundamentals with hands-on experience and real-world relevance. We continuously strengthen our curriculum, teaching practices and quality processes so that our programmes meet global standards. Experiential learning, international collaborations, and meaningful engagement with industry help our students develop the skills, confidence, and values needed to succeed in a fast-changing world. Equally, we encourage our faculty and researchers to pursue work that is rigorous, collaborative, and useful to society. I am proud of the Thapar community's spirit of excellence, and I encourage everyone to keep learning, innovating, and taking the Institute to greater heights.

ABOUT THE SPEAKER

Prof. Ajay Batish is the Pro Vice-Chancellor of Thapar Institute of Engineering and Technology (TIET), Patiala, and a Professor of Mechanical Engineering. He earned his PhD from Thapar University and, before joining academia, spent nearly a decade heading Quality Assurance in industry. He spearheaded the development of the Experiential Learning Centre in collaboration with the University of Waterloo. His research spans advanced manufacturing, process optimization, ergonomics, and quality engineering, with numerous publications and projects funded by DST, DRDO, UGC, AICTE, and Industry.

MESSAGE FROM ORGANIZING COMMITTEE



Shri. Alok Mukherjee
President, TRS
Patron

MESSAGE

On behalf of The Robotics Society (TRS), India, I warmly welcome all delegates, experts, and participants to the Two-Day International Workshop on “Intelligent Systems: The Future of Robotics and AI.” We stand at a critical inflection point where artificial intelligence and advanced robotics are redefining human capability, as robotics moves beyond industrial automation into autonomous air, ground, surface, and underwater vehicles, adaptive medical systems, and intelligent edge machines. This workshop provides an excellent platform to bridge foundational research with real-world deployment. Over the next two days, I encourage you to challenge existing paradigms, exchange interdisciplinary ideas and cultivate collaborations that extend well beyond this venue. I commend the organizing team for curating this timely forum, and may these two days spark transformative ideas.

ABOUT THE SPEAKER

Shri. Alok Mukherjee is a distinguished defence scientist, technological innovator and leader in robotics and unmanned systems in India. With over three decades at the Defence Research and Development Organisation (DRDO), he served in pivotal leadership roles, including Group Director – Robotics at the Research & Development Establishment (Engineers), Pune. His work includes the UAV ‘NETRA’ and the ROV ‘Daksh’, the first indigenous military robot for bomb and IED handling. As President of TRS, India, he drives national robotics research, industry–academia collaboration, and student mentoring. He is also the founder of i4 Marine Technologies Pvt. Ltd, a Pune-based startup in marine robotics.

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Inderveer Chana
Joint Director, DSAI, TIET Patiala
Workshop Chair

MESSAGE

I extend my warm welcome to all attendees of the International Workshop on Intelligent Systems. The TIET–UQ Centre of Excellence in Data Science and AI is committed to fostering impactful collaborations with startups, industries, and public institutions to co-develop AI tools that benefit society. Our work is organised around key thematic areas, including health and medicine, energy and environment, cybersecurity, smart agriculture, sustainable infrastructure, and behavioural science. Through seed funding for collaborative projects, travel scholarships, and industry roundtables, we aim to build a vibrant community of researchers. I invite students, faculty members, and industry partners to join us in co-creating innovative solutions that make a meaningful difference to people, the economy, and the planet.

ABOUT THE SPEAKER

Prof. Inderveer Chana is a Professor in the Department of Computer Science and Engineering at Thapar Institute of Engineering and Technology (TIET), Patiala, and the Joint Director of the TIET–UQ Centre of Excellence in Data Science and AI. She holds a PhD in Computer Science with specialization in grid computing and an M.E. in Software Engineering from TIET, and has served as Dean of Student Affairs. Her research interests include cloud computing, energy-aware computing, machine learning, and healthcare informatics. She has published widely in venues such as ACM Computing Surveys and IEEE Transactions on Cloud Computing, supervised many PhD theses, and led projects sponsored by UGC, AICTE, CSIR, DST, and ICMR.

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Tim Miller
Joint Director, DSAI, TIET Patiala
Workshop Chair

MESSAGE

On behalf of the Centre of Excellence in Data Science and AI, I extend my warm greetings to all participants of the workshop. Why do advanced autonomous agents and robots still fail to collaborate well with people? Building better human-AI and human-robot systems requires more than computing and mechatronics. It requires us to understand how users of these tools want and need to interact with them. In this workshop, we turn to research on how humans interact with each other, and use insights from this research to build human-AI and human-robot systems. Using explainable artificial intelligence as a primary case study, we demonstrate how models of human-to-human explanation can be operationalised to build explainable AI systems that users can more intuitively understand, trust, and use.

ABOUT THE SPEAKER

Prof. Tim Miller is the TIET-UQ Chair of Data Science at TIET and Professor of Artificial Intelligence in the School of Electrical Engineering and Computer Science at The University of Queensland. He is also the Joint Director of the Centre of Excellence. His mission is to improve the capabilities of expert decision makers by using artificial intelligence to do the hard things. Tim's research draws on machine learning, reinforcement learning, AI planning, interaction design, and cognitive science to help people make better decisions. He has done work in areas including explainable AI, human-AI mixed-initiative planning, and human-centered decision support.

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Ashish Singla
Professor, TIET Patiala
Convener

MESSAGE

The workshop aims to bring together eminent academicians, researchers, industry leaders, and students to explore emerging developments in Robotics, Artificial Intelligence, and Intelligent Systems. With distinguished national and international speakers, interactive technical sessions, and industry-led hands-on activities, the event will provide a vibrant platform for knowledge exchange and innovation, bridging the gap between academic research and real-world applications. We hope it will inspire new ideas, foster interdisciplinary collaborations, and encourage the next generation of researchers and innovators. I warmly welcome all participants, speakers, industry partners, and students to these two exciting days of learning, interaction, and collaboration.

ABOUT THE SPEAKER

Prof. Ashish Singla is a Professor in the Department of Mechanical Engineering at Thapar Institute of Engineering and Technology (TIET), Patiala, with more than two decades of experience in industry, teaching and research. He received his PhD in flexible robotic manipulators from IIT Kanpur. His research spans flexible, underactuated and redundant manipulators, real-time dynamics and control, exoskeletons, and assistive robotic devices. He has been a Visiting Faculty at SUNY Stony Brook, USA, and the University of Gävle, Sweden, and served as Programme Co-Chair of Advances in Robotics (AIR-2023) at IIT Ropar. He has published 100+ SCI journal and conference papers and led many sponsored projects funded by agencies including DST-SERB and DRDO. As President of the Mechatronics and Robotics Society (MARS), TIET, he takes pride in leading a team that secured the Champions title at the International Rover Challenge (IRC) 2026 and 1st Runner-Up at the International Space Drone Challenge (ISDC) 2026.

MESSAGE FROM ORGANIZING COMMITTEE



Prof. Sachin Kansal
Associate Professor, TIET Patiala
Co-Convener

MESSAGE

Robotics and Artificial Intelligence are no longer confined to laboratories or theoretical explorations; they are increasingly becoming integral to how we perceive, interact with, and shape the physical world. As we move towards an era of intelligent and autonomous systems, the real challenge lies not merely in developing smarter algorithms, but in translating intelligence into meaningful actions, reliable systems, and impactful solutions. I believe the future of intelligent systems will be shaped by our ability to integrate mechanical intelligence, computational methods, data-driven learning, and human-centric design, and to prepare young researchers to develop technologies that are intelligent, accessible, responsible, and relevant to societal needs.

ABOUT THE SPEAKER

Prof. Sachin Kansal is an Associate Professor in the Department of Computer Science and Engineering at Thapar Institute of Engineering and Technology (TIET), Patiala, and Coordinator (Research and Innovation) of the TIET–UQ Centre of Excellence in Data Science and AI. He received his PhD in Robotics from IIT Delhi and his M.Tech from IIIT Allahabad, and was a Postdoctoral Research Fellow in Robotics and Computer Vision at the University of Birmingham, UK. His research covers robotics, computer vision, mechatronics, human-robot interaction, and deep learning.

SPEAKER



Prof. Ahmed Chemori

Senior CNRS Researcher
LIRMM • University of Montpellier, France

TALK TITLE

When Control Design Serves Human Assistance and Rehabilitation with Exoskeletons

ABSTRACT

Assisting older adults and people with disabilities in their daily activities through wearable robotic technologies has attracted growing interest, driven by the increasing prevalence of mobility impairments and the rapid aging of the global population. Advances in compact components, low-power electronics, lightweight actuators, and accessible sensing are making assistive devices more wearable, autonomous and affordable, and exoskeletons, wearable biosensors and functional electrical stimulation systems are steadily moving from research prototypes to practical applications. In this context, control plays a central role in enabling wearable robots to interact safely, intuitively, and effectively with the human body. This talk reviews major challenges and recent advances in wearable robotics, with a focus on control strategies for assistive and rehabilitation exoskeletons, illustrated through numerical simulations and real-time experiments involving human subjects.

ABOUT THE SPEAKER

Prof. Ahmed Chemori received his M.Sc. and PhD degrees in Automatic Control from the Polytechnic Institute of Grenoble, France, in 2001 and 2005. He is currently a Senior CNRS Researcher in Automatic Control and Robotics at the LIRMM laboratory, University of Montpellier. His research focuses on nonlinear control, including adaptive, robust and predictive control, with applications to wearable, parallel and marine robotics. He has authored or co-authored more than 200 scientific publications and has co-supervised 28 PhD theses and over 40 M.Sc. theses. He leads the Marine Robotics Team and chairs the LIRMM Valorization Committee. He serves as Technical Editor for IEEE/ASME Transactions on Mechatronics, is an IEEE Senior Member, and is a member of several IFAC Technical Committees.

SPEAKER



Prof. Ashish Dutta

Professor (HAG)
Department of Mechanical Engineering
Indian Institute of Technology Kanpur, India
Vice-President, TRS

TALK TITLE

BCI Based Hand Exoskeleton for Neurorehabilitation of Stroke Patients

ABSTRACT

Loss of hand function after stroke or spinal cord injury has driven rapid growth in wearable assistive devices. Stroke survivors recover upper-limb movement far better when physical practice is combined with motor imagery practice of daily activities. This talk presents the design of two-finger and three-finger exoskeletons based on natural finger motion trajectories, enabling simple grasping as well as fine manipulation of small objects. A novel brain–computer interface fusing EMG and EEG signals controls the exoskeleton and provides visual neurofeedback to keep therapy focused. The talk concludes with results of a six-week clinical trial on eight hemiparetic stroke patients in India and the UK, all of whom showed positive rehabilitative outcomes.

ABOUT THE SPEAKER

Prof. Ashish Dutta obtained his PhD in Systems Engineering from Akita University, Japan, M.Tech from Jadavpur University and B.Tech from NIT Calicut. He worked at the Bhabha Atomic Research Centre from 1994 to 2000 on tele-manipulator design for nuclear applications. Since 2002 he has been with the Department of Mechanical Engineering at IIT Kanpur, where he is a Professor (HAG) and Head of the Department (2023–2026). He is also a visiting professor at Kyushu Institute of Technology, Japan. His research spans humanoid robotics, intelligent control and rehabilitation engineering. He is the Vice-President of The Robotics Society of India and has served on the advisory boards of the ISRO Chandrayaan-II and Gaganyaan missions.

SPEAKER



Prof. Shyamanta M. Hazarika

DST-INAE Abdul Kalam Technology Innovation National Fellow
Professor, Department of Mechanical Engineering
Indian Institute of Technology, Guwahati, India

TALK TITLE

Understanding Human Cognition: From Object Affordance to Brain Networks

ABSTRACT

Object affordance – the action possibilities that objects offer to an observer – links perception, cognition, and action. This talk presents EEG-based investigations into object affordance, action observation, motor imagery, and affordance-driven action-language comprehension, with emphasis on how affordance congruency modulates neural responses. Dynamic Causal Modelling and Bayesian model comparison are used to characterise the network-level mechanisms involved, revealing premotor–parietal interactions in the effective connectivity architecture. The talk concludes by connecting these findings to human-centred intelligent systems, showing how affordance representation and perception–action integration bridge computational neuroscience and the design of intelligent machines, and underpin work on human–robot interaction and wearable robotics for neurorehabilitation.

ABOUT THE SPEAKER

Prof. Shyamanta Moni Hazarika is a DST-INAE Abdul Kalam Technology Innovation National Fellow. He is a Professor of Mechanical Engineering and of the Mehta Family School of Data Science and AI at IIT Guwahati, where he leads the Biomimetic Robotics and Artificial Intelligence Lab. His research focuses on bionic prosthetics and robotic neurorehabilitation, integrating AI and machine learning with robotics and neuroscience to improve the lives of individuals with neuromotor disabilities. He previously served as Professor of Computer Science and Engineering at Tezpur University and as a substitute Professor at the University of Bremen, Germany. He holds a B.E. from Assam Engineering College, an M.Tech. in Robotics from IIT Kanpur and a PhD in Artificial Intelligence from the University of Leeds, England.

SPEAKER



Prof. Ekta Singla

Director, NAMTech School of Robotics
Secretary, TRS

TALK TITLE

Distinct Features of Robotics Industry: A National Context

ABSTRACT

The Indian robotics industry is undergoing a structural transition from automation consumption to indigenous product innovation across manufacturing, healthcare, defence, agri-tech and logistics - from an automation user to a creator of Indian-origin robots. While the emerging market promises immense employability, a critical gap persists between conventional robotics education and the fast-evolving needs of industry. This session highlights the paradigm shift required in learning, from component-based theory to a systems approach centred on Physical-AI, supported by shared laboratory infrastructure, industry-academia co-creation and a curriculum built around real-world systems. The discussion will delineate the employability contours of the emerging market and propose a cohesive model to align research, pedagogy and Indian-origin robotics innovation.

ABOUT THE SPEAKER

Prof. Ekta Singla is the Director of NAMTech School of Robotics, an education initiative from ArcelorMittal in Gujarat, and Secretary of The Robotics Society (TRS) India. A former Head of the Mechanical Engineering Department at IIT Ropar, she holds a PhD in Robotics from IIT Kanpur and has over two decades of academic and research experience, including postdoctoral work at TU Berlin and UPMC Paris and a visiting professorship at SUNY Stony Brook. Her research spans modular and reconfigurable robotics, healthcare and exoskeleton systems, and robotic assistance in vertical farming, and wearable exoskeletons. A member of the CII National Committee on Robotics and Automation, she contributes to national policy discussions on robotics education. Her accolades include a national award, a Director's gold medal and fellowships from DFG Germany and CNRS France. She has served on the Governing Council of The Robotics Society since 2021 and founded the PUNJRobotics platform at IIT Ropar in 2014.

SPEAKER



Dr. Rajesh Kumar

Principal Scientist
Head, Advanced Robotics
Addverb

TALK TITLE

Physical AI: Infrastructure and Cognition

ABSTRACT

Robots are evolving from executing predefined routines to perceiving, learning, and adapting to changing environments. This talk explores that transition through manipulation and locomotion intelligence, highlighting how reinforcement learning, vision-language-action models and world models work alongside planning and control. It examines the scale, diversity and quality of data needed to generalize across tasks and environments. Equally important are robot hardware, actuators, sensors, and batteries, which determine physical capability, perception and endurance. Drawing on industrial robotics, the talk discusses how intelligence, data, and hardware must advance together to enable adaptable, reliable autonomy.

ABOUT THE SPEAKER

Dr. Rajesh Kumar is a Principal Scientist at Addverb, where he leads the development of advanced robotic platforms across hardware, software and AI. His technical expertise spans robot kinematics and dynamics, nonlinear and model predictive control, trajectory optimization, visual and LiDAR SLAM, reinforcement learning, vision-language-action models, and world models. His work covers manipulation, legged locomotion, autonomous navigation, teleoperation, robot control and cognition, integrating algorithms with actuators, sensing, and real-time robot software. He leads development across cobots, quadrupeds, dual-arm systems, and humanoids.

SPEAKER



Dr. Madan Dabbeeru

Founder, Eizen AI (USA, UAE & INDIA)
AI Research & Engineering

TALK TITLE

Evolving the Watchers: Verifier-Driven Perpetual Learning for Robot Observability

ABSTRACT

As robots move from controlled pilots to fleet-scale deployments, a new problem emerges: how do we reliably know whether a robot is performing as intended? Telemetry reports what the machine believes it did, and often misses slow degradation, subtle grasp failures and timing drift, while observer models trained once decay as sites and products change. This talk presents Eizen AI's approach to robot observability as a continuous discipline, RobOps, built on a dual-vision pairing of the robot's ego-view with an external exo-view. Observer models are evolved in a simulated environment with fault injection, where failure scenarios and observers co-evolve under a panel of independent verifiers. The talk closes with open research challenges and opportunities for student collaboration.

ABOUT THE SPEAKER

Dr. Madan Dabbeeru is the Founder of Eizen AI, with over 22 years of AI research and engineering across the USA, Dubai and Bangalore. He has led AI initiatives at Rakuten India as Head of Research and was a key architect at the unicorn startup Phenom People, delivering AI across defense, healthcare, banking, and e-commerce. His work includes unmanned vehicles for the US Navy and Air Force, AI for cancer research and AI that generates AI. He holds a Post-Doc and PhD in Cognitive Robotics from the University of Maryland, IIT Kanpur and BITS Pilani.

SPEAKER



Prof. Harish PM

Professor, IIT Gandhinagar
Director, AB6 Robotics

TALK TITLE

Intelligence Through Interaction: Sensing, Learning, and Control for Robotic Manipulation

ABSTRACT

Robots that open a jammed bottle lid or snap-fit a spectacle lens into its frame need far more than precise motion; they need a kind of intelligence that senses, perceives and learns through real-world interaction. This talk explores how physical interaction can be designed to understand and complete manipulation tasks, drawing on recent work from our lab: coordinating a delicate assembly, dismantling an unfamiliar mechanism, and combining vision and contact control to wipe multi-panel surfaces. These examples expose a basic challenge: the same action can lead to different outcomes in the real world, making sensing and learning from physical interactions crucial for robot skills that respond to uncertainty in contact-rich manipulation.

ABOUT THE SPEAKER

Prof. Harish J. Palanthandalam-Madapusi received his B.E. from SPCE, University of Mumbai, and his M.S.E. and PhD from the University of Michigan in 2007. He was an Assistant Professor at Syracuse University until 2011, when he joined IIT Gandhinagar, where he has held several leadership roles, including Dean (Campus Development), Dean (Student Affairs) and Dean (General Administration). He has authored more than 90 journal and proceedings papers and led over twenty research and consulting projects with agencies such as ISRO and the Indian Navy and with industry partners. He is co-founder of AB6 Robotics, and his research focuses on robotics and robot control, particularly manipulation.

SPEAKER



Prof. Santhakumar Mohan

Professor
Department of Mechanical Engineering
Indian Institute of Technology, Palakkad, India

TALK TITLE

Intelligent Manufacturing and Industry 5.0 (The Role of Mobile Manipulators)

ABSTRACT

Industry 5.0 emphasizes human-centric, sustainable and resilient manufacturing by integrating intelligent automation with human expertise. In this context, mobile manipulators, which combine the mobility of robotic platforms with the dexterity of robotic arms, offer significant potential for flexible and adaptive manufacturing systems, performing material handling, machine tending, assembly, inspection and intralogistics across dynamic environments. Integrating artificial intelligence, machine vision, force sensing and intelligent motion planning lets them operate autonomously while collaborating safely with human workers. The talk explores the enabling technologies, emerging applications and key challenges in deploying mobile manipulators for intelligent manufacturing.

ABOUT THE SPEAKER

Prof. Santhakumar Mohan is a Professor in the Department of Mechanical Engineering and Head of the Centre for Robotics and Control at IIT Palakkad. He received his PhD in Robotics and Control from IIT Madras in 2010 and has held visiting positions at IISc Bangalore, KAIST, RWTH Aachen, ECN France and BSTU Russia. His honours include the Outstanding Young Scientist Award from the Korea Robotics Society and the Alexander von Humboldt Fellowship. He has published more than 150 research papers, holds 13 granted Indian patents, and has served as Technical Editor of IEEE/ASME Transactions on Mechatronics and Associate Editor of IEEE Robotics and Automation Letters. He is the founding Director of IPTIF, IIT Palakkad TRIP and TECHIN.

SPEAKER



Prof. Nirav Patel

Associate Professor
Department of Engineering Design,
Indian Institute of Technology Madras, India

TALK TITLE

Robotics and Autonomy in Minimally Invasive Surgery

ABSTRACT

Minimally invasive surgery (MIS) has transformed patient outcomes, but it places significant demands on the surgeon: restricted access, limited visual feedback, reliance on image guidance and interaction with soft, deforming tissue. Robotics can improve dexterity and precision and, increasingly, take on parts of the surgical workflow autonomously. This talk presents our work from robotic assistance to autonomous interventions: a CT-guided robotic system for accurate instrument placement in neurosurgery; AutoPlan, which performs fully autonomous trajectory planning for steerable needles; autonomous endoscope positioning by a cobot in laparoscopic surgery; and the open problem of manipulating deformable objects, using physics-based models with learned dynamics and planning.

ABOUT THE SPEAKER

Prof. Nirav Patel is an Associate Professor in the Department of Engineering Design and leads the INSPIRE Lab at IIT Madras. He was a postdoctoral research fellow at the Laboratory for Computational Sensing and Robotics, Johns Hopkins University, from 2017 to 2020, and received his PhD in Robotics Engineering from Worcester Polytechnic Institute in 2017. He previously worked on industrial robots at ABB's corporate research center in Bangalore. His research covers image-guided robotic systems for percutaneous intervention, surgical and motion planning, sensors and control strategies for safe robot-assisted minimally invasive surgery, with a particular interest in clinical translation.

SPEAKER



Dr. Pranav Lad

Senior Customer Success Engineer
Education Team
MathWorks, Pune, India

TALK TITLE

Reimagining Robot Development with AI and Agentic Workflows

ABSTRACT

Artificial intelligence is reshaping robotics beyond perception and decision-making by bringing intelligence to the entire development lifecycle. This talk explores how AI can support key stages of robot development, from platform design and sensor integration to perception, planning and control, and how combining AI with simulation and model-based engineering can shorten development cycles, and improve design exploration. It also introduces Agentic AI as an emerging approach for orchestrating end-to-end engineering workflows, with examples using the MATLAB and Simulink Agentic Toolkits showing how AI agents can interact with models, simulations, and tools while supporting human-guided workflows, contributing to the emerging vision of Physical AI.

ABOUT THE SPEAKER

Dr. Pranav Lad is a Senior Customer Success Engineer at MathWorks, where he has been a member of the Education Team in Pune, India, for over six years, collaborating closely with academic institutions to accelerate research and enhance learning outcomes. His expertise spans robotics, autonomous systems, vibrations, nonlinear dynamics, and artificial intelligence, with a current focus on the modelling and simulation of robotics and autonomous systems, especially UAVs and automated driving. He holds a PhD and M.Tech. in Mechanical Engineering from IIT Bombay, where he also worked as a research assistant on industry consulting projects, and a bachelor’s degree from the University of Pune.

SPEAKER



Prof. Sitikantha Roy

Joint Faculty, Department of Applied Mechanics
School of Artificial Intelligence
Indian Institute of Technology, Delhi

TALK TITLE

Biomechanics in Human Digital Twin and Wearable Robotic System Design

ABSTRACT

Advances in biomechanics are transforming our ability to understand, predict and enhance human movement, health, and performance. This talk presents recent research at the intersection of multi-scale computational biomechanics, with predictive healthcare examples, and human-centered engineering through wearable robotic technologies. By combining physics-based modeling, scientific AI, and experimental human studies, biomechanical frameworks can predict injury risk, characterize disease progression, and optimize personalized interventions. The talk also highlights the development of wearable robotic systems, including soft exosuits and assistive technologies, with emphasis on human-in-the-loop modeling, neuromuscular dynamics, and digital-twin approaches that enable personalized prediction and adaptive control.

ABOUT THE SPEAKER

Prof. Sitikantha Roy is a Fulbright-Nehru Academic & Professional Excellence Award winner and a joint faculty member in the Department of Applied Mechanics and the School of Artificial Intelligence at IIT Delhi, where he heads the Biomechanics and Soft Robotics Laboratory. His research focuses on human-centered dynamical systems, scientific-AI-driven hybrid predictive modeling, computational biomechanics, wearable robotics, and soft robotic systems. His work aims to bridge fundamental mechanics, computation, and experimentation to develop predictive technologies for healthcare, rehabilitation, and human augmentation.

SPEAKER



Prof. Hari Kumar Voruganti

Professor, Department of Mechanical Engineering
National Institute of Technology Warangal, India

TALK TITLE

Evolutionary Pareto Optimization for Coverage Path Planning in Industrial Mobile Robotics

ABSTRACT

Efficient coverage path planning is essential for mobile robots operating in complex real-world environments, where missions demand a delicate balance among competing performance metrics such as energy consumption, turning frequency, and elevation constraints. This talk explores advanced multi-objective evolutionary frameworks designed to generate optimal Pareto fronts, moving beyond single-metric approaches to provide robust, adaptable navigation strategies. It shows how integrating these optimization methods translates into substantial efficiency gains and smoother physical execution for autonomous mobile robotic systems.

ABOUT THE SPEAKER

Prof. Hari Kumar Voruganti is a Professor in the Department of Mechanical Engineering at the National Institute of Technology Warangal (NITW), and holds a PhD from IIT Kanpur. His research focuses on multi-DOF robot kinematics, redundancy resolution, and collision-free path planning in complex workspaces. He actively explores optimization-driven methodologies for high-performance automation and robotic navigation. His recent work addresses coverage path planning of mobile robots in uneven terrains.

SPEAKER



Dr. Anubhav Dogra

Senior Autonomy Engineer
Controls & AI
Humanoid, London, UK

TALK TITLE

From Demonstrations to Deployment: Integrating AI, Control and Hardware in Wheeled Humanoids

ABSTRACT

Wheeled humanoids bring mobile manipulation to environments designed for people, but the full potential of a learned policy depends on a well-integrated system. This talk examines that integration challenge through the development and deployment of wheeled humanoid systems. Following a representative manipulation task, it traces the path from teleoperation and data collection through imitation and reinforcement learning to execution on hardware. It explores how actuator integration, kinematics, motion planning, compliant control, simulation, calibration, and fault handling make learned behavior physically feasible, responsive and repeatable, and concludes with practical considerations for safe physical interaction, consistent behavior across robots, and systematic evaluation.

ABOUT THE SPEAKER

Dr. Anubhav Dogra is a Senior Autonomy Engineer in Controls & AI at Humanoid in London, UK. He has a background in mechanical engineering, a PhD in robotics from IIT Ropar and postdoctoral research experience at the University of Warwick, UK. His work focuses on integrating software with physical hardware, bridging learned AI policies and control systems in wheeled and bipedal humanoids. His work spans robot bring-up, actuator and sensor integration, whole-body kinematics and dynamics, simulation development, and controllers that ensure safe, repeatable physical interaction in industrial deployments.

SPEAKER



Mr. Muralikrishnan E

Head – Edutech Future Mobility Labs
Edutech India Pvt. Ltd.

TALK TITLE

From Wheels to Wings: How AI Is Redefining Autonomous Robotics

ABSTRACT

Autonomous robotics is rapidly moving beyond isolated machines toward intelligent systems that can perceive, decide, navigate, and act in real-world environments. This talk explores the common intelligence stack behind both autonomous ground vehicles and aerial robotic systems, examining how AI, computer vision, LiDAR, sensor fusion, localization, path planning and control come together to enable safe and adaptive autonomy. Using ground and aerial mobility as complementary examples, it also discusses the transition from simulation to physical systems, the role of edge AI and real-time decision-making, and research opportunities in robotics, AI, multi-agent systems, and Physical AI.

ABOUT THE SPEAKER

Mr. Muralikrishnan E is the Head of Edutech Future Mobility Labs at Edutech India Pvt. Ltd., where he leads initiatives in autonomous mobility, ADAS, electric vehicles, software-defined vehicles and intelligent robotic systems. His interests include autonomous ground and aerial platforms, sensor fusion, perception, localization, path planning, control, ROS 2/Autoware, and Physical AI. He works closely with academic institutions, research organizations and industry to develop future-mobility laboratories, research platforms, and training programs. He holds an M.S. in Telecommunications Engineering from Victoria University, Melbourne, and a B.E. in Electronics and Communication Engineering from the University of Madras.

SPEAKER



Mr. Saurabh Joshi

Regional Account Manager
Noraxon USA Inc.

HANDS-ON SESSION

Multi-Device Biomechanics for Robotics and Assistive Technologies

ABSTRACT

This 2½-hour hands-on session demonstrates how surface EMG, wearable 3D motion capture, force measurements, pressure insoles, and synchronized video can be integrated within Noraxon's myoRESEARCH platform to simplify data collection for human movement research. Through live demonstrations, participants will examine muscle activity, movement, and loading together during the same task. The session covers device integration and synchronization, connecting compatible third-party equipment, and data export for computational biomechanics and control research, with applications in human–robot interaction, wearable robotics, exoskeleton assessment, prosthetics, and orthotics.

ABOUT THE SPEAKER

Mr. Saurabh Joshi is Regional Account Manager at Noraxon USA Inc., supporting researchers and institutions across India. He brings around 14 years of experience in scientific measurement technologies and lab solutions, including technical support, product training and work with biomechanics, and rehabilitation equipment. He works closely with academic, clinical and industry teams, helping institutions assess their requirements, plan laboratories, and make effective use of their equipment. At Noraxon, his responsibilities include business development, institutional relationships, and coordination with local partners. He holds an MBA in Analytics and Business Development.

SPEAKER



Mr. Puneet Jindal

Founder & CEO
Labellerr AI and Robotics Data

HANDS-ON SESSION

The Missing Layer in Robotics: Real-World Data Supply Chains, with Lessons from Indian Factories

ABSTRACT

Progress in robot learning is increasingly limited by data rather than model architecture. This session walks through the full data supply chain for physical AI: capturing egocentric multi-modal video in working factories, consent and privacy handling, quality control, multi-layer annotation (hands, objects, actions, language), dataset formats for training, and the move from human video to robot policies via retargeting and teleoperation. Drawing on field deployments across Punjab manufacturing units, it shows what works, what breaks, and where open research problems remain, such as embodiment gap, label noise, and evaluation. Attendees will see each stage visually and leave with a practical checklist for designing their own robot-data pipelines.

ABOUT THE SPEAKER

Mr. Puneet Jindal is Founder & CEO of Labellerr AI and Robotics Data. He holds a B.E. in Computer Science from Thapar Institute and completed a Data Science and AI programme at the Indian School of Business. He spent nine years building AI software and data products at Walmart, Delhivery, RateGain and startups across logistics, e-commerce, travel, and healthcare. He founded Labellerr AI, a data annotation platform used by research groups and companies including Stanford, MIT, Toyota Research Institute, and Volvo. In 2026, he launched Robotics Data to build the real-world data collection layer for robot learning, and has collected about 5,000 hours of egocentric data on Indian and global factory floors.

SPEAKER



Mr. Rohit Ranjan Bhattacharjee

Customer Success Engineer
Education Team
MathWorks, India

HANDS-ON SESSION

Comprehensive Robotic Systems Development - Simulation, Control and Hardware Integration

ABSTRACT

This hands-on workshop introduces an end-to-end workflow for developing an autonomous robotic system, starting from model import and controls, through interfacing with ROS, to hardware deployment, using MATLAB, Simulink, and Simscape. Participants will learn to create robot models from CAD and URDF files, apply PID and reinforcement learning-based control, connect MATLAB and Simulink with ROS/ROS 2, perform co-simulation in virtual environments, and explore hardware integration and model validation through examples involving robotic manipulators and mobile robots.

ABOUT THE SPEAKER

Mr. Rohit Ranjan Bhattacharjee is a Customer Success Engineer with the Education Team at MathWorks. He is completing his PhD in Aerospace Engineering at IIT Madras, where he investigated flows in gas turbine engines using advanced image processing techniques. He has over seven years of experience in academic research and has published several papers in reputed journals, including one that received the Distinguished Paper Award from the Proceedings of the Combustion Institute. Previously, he earned a Master's degree by research at the University of Ottawa, where he studied supersonic flows, and a Bachelor's degree in Aerospace Engineering from Carleton University, Canada.

TRS-DSAI Workshop 2026 – Programme Schedule

Friday, 30 October – Saturday, 31 October 2026

Day	09:00 – 10:00 AM	10:00 – 10:45 AM	11:00 – 11:30 AM	11:30 AM – 12:15 PM	12:15 – 01:00 PM	01:00 – 02:00 PM	02:00 – 02:45 PM	02:45 – 03:30 PM	03:30 – 04:00 PM	04:00 – 04:15 PM	04:15 – 05:00 PM	05:00 – 06:00 PM	06:00 – 08:00 PM
Fri 30 Oct 2026	Intelligent Systems (Auditorium / LP-108)						Robotics in Industry (LP-108)						
	Inaugural Session	Dr. Rajesh Kumar (Addverb)	**High Tea**	Prof. Ashish Dutta (IIT Kanpur)	Prof. Harish PM (IIT Gandhinagar)	**Lunch Break**	Dr. Anubhav Dogra (Humanoid, UK)	Prof. S K Mohan (IIT Palakkad)	Mr. Muralikrishnan E (Edutech India)	**Tea Break**	Prof. Hari Kumar (NITW)	Prof. Ekta Singla TRS GBM	
											HOS 1: MathWorks (Mr. Rohit Bhattacharjee)		
Sat 31 Oct 2026	AI/ML in Robotics (LP-108)						Robotics in Healthcare (LP-108)						
	Dr. Madan Dabbeeru (Eizen, USA)	Prof. S M Hazarika (IIT Guwahati)	**Tea Break**	Prof. Tim Miller (UQ, AUS)	Dr. Pranav Lad (MathWorks)	**Lunch Break**	Prof. Ahmed Chemori (LIRMM, France)	Prof. Sitikantha Roy (IIT Delhi)	Mr. Aditya Marathe (Nugenix)	**Tea Break**	Prof. Nirav Patel (IIT Madras)	**Valedictory Session**	
							HOS 2: Noraxon (Mr. Saurabh Joshi)			**HOS 3: Labellerr AI (Mr. Puneet Jindal)**			

HOS : Hands-on Sessions (LP-109)

Group Photograph : 10:45 - 11:00 am

Day 1 Schedule (Friday, 30 October 2026)		
08:00 AM - 09:00 AM	Registration	
09:00 AM - 10:00 AM	Inaugural Session	
Theme 1 - Intelligent Systems (Auditorium / LP-108)		
10:00 AM - 10:45 AM	Dr. Rajesh Kumar (Addverb)	
10:45 AM - 11:00 AM	Group Photograph	
11:00 AM - 11:30 AM	High Tea	
11:30 AM – 12:15 PM	Prof. Ashish Dutta (IIT Kanpur)	
12:15 PM – 01:00 PM	Prof. Harish PM (IIT Gandhinagar)	
01:00 PM – 02:00 PM	Lunch Break	
Theme 2 - Robotics in Industry (LP-108)		
02:00 PM – 02:45 PM	Dr. Anubhav Dogra (Humanoid, UK)	
02:45 PM – 03:30 PM	Prof. Santhakumar Mohan (IIT Palakkad)	
03:30 PM – 04:00 PM	Mr. Muralikrishnan E (Edutech India)	
04:00 PM – 04:15 PM	Tea/Coffee Break	
04:15 PM – 05:00 PM	Prof. V. Hari Kumar (NITW)	HOS 1: MathWorks Mr. Rohit Ranjan Bhattacharjee (LP-109)
05:00 PM – 06:00 PM	Prof. Ekta Singla, TRS GBM (TRS Secretary)	
06:00 PM – 08:00 PM		

Day 2 Schedule (Saturday, 31 October 2026)

Theme 3 - AI/ML in Robotics (LP-108)		
09:00 AM - 10:00 AM	Dr. Madan Dabbeeru (Eizen, USA)	
10:00 AM - 10:45 AM	Prof. S M Hazarika (IIT Guwahati)	
10:45 AM - 11:00 AM	Group Photograph	
11:00 AM - 11:30 AM	Tea Break	
11:30 AM – 12:15 PM	Prof. Tim Miller (UQ, AUS)	
12:15 PM – 01:00 PM	Dr. Pranav Lad (MathWorks)	
01:00 PM – 02:00 PM	Lunch Break	
Theme 4 - Robotics in Healthcare (LP-108)		
02:00 PM – 02:45 PM	Prof. Ahmed Chemori (LIRMM, France)	HOS 2: Noraxon Mr. Saurabh Joshi (LP-109)
02:45 PM – 03:30 PM	Prof. Sitikantha Roy (IIT Delhi)	
03:30 PM – 04:00 PM	Mr. Aditya Marathe (Nugenix)	
04:00 PM – 04:15 PM	Tea Break	HOS 3: Labellerr AI Mr. Puneet Jindal (LP-109)
04:15 PM – 05:00 PM	Prof. Nirav Patel (IIT Madras)	
05:00 PM – 06:00 PM	**Valedictory Session**	

Schedule

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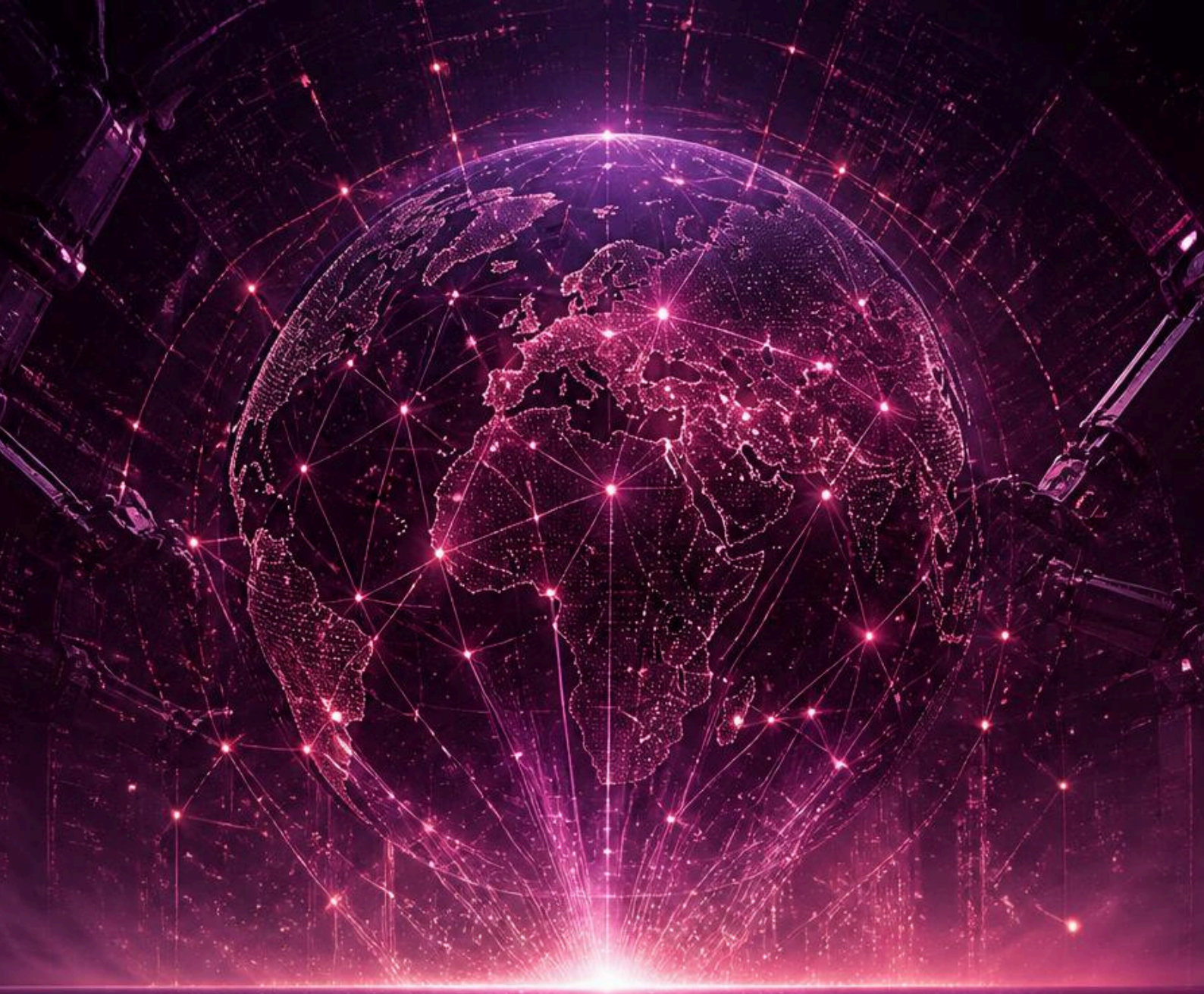
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